

HAT-P-36b

R-1774

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-36_180124 · created 2026-08-02T22:28:28+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2458142.8932 +/- 0.0016 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1392 +/- 0.0087
Transit depth (Rp/Rs)^2	1.94 +/- 0.24 [%]
Orbital Inclination (inc)	84.84 +/- 2.18
Airmass coefficient 1 (a1)	1.008 +/- 0.011
Airmass coefficient 2 (a2)	-0.0077 +/- 0.0052
Scatter in the residuals of the lightcurve fit is	1.07 %
Best Comparison Star	#1 - [175, 138]
Optimal Aperture	4.53
Optimal Annulus	11.86
Transit Duration (day)	0.092 +/- 0.0056

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1392 $\pm$ 0.0087 vs 0.126 (deviation 1.52 σ)
Duration fit vs published	0.092 d vs 0.0925 d (deviation 0.09 σ)
Mid-time O-C	5.6 min
Depth SNR	16.0
Residual scatter	1.07 %

Light curve

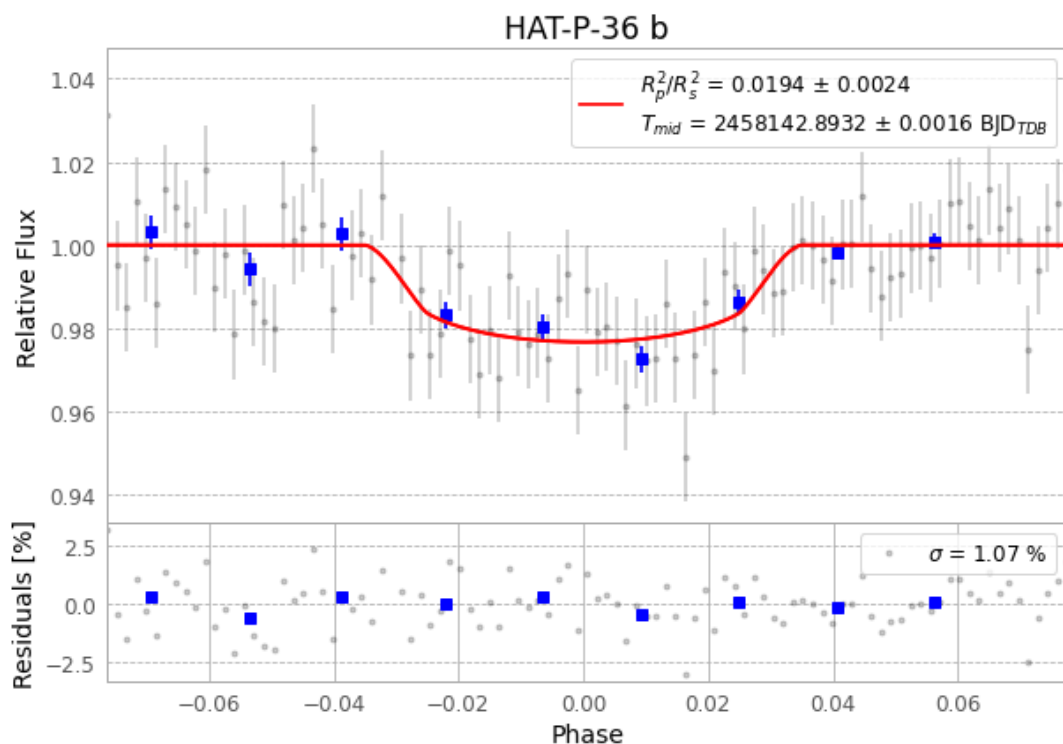


Figure 1 — FinalLightCurve_HAT-P-36 b_2018-01-23.png (SHA-256 207b6a63aa03d122...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [447, 202], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 9cf2ce01e7e17841...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

99 FITS frames (65 MB), HATP-36180124065412.FITS → HATP-36180124114812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-180124.log (3b1e4fecaa49...), FinalLightCurve_HAT-P-36 b_2018-01-23.png (207b6a63aa03...), FinalLightCurve_HAT-P-36 b_2018-01-23.csv (ddc82a9cc05b...)

Compute resources

Wall time 165.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 22:28Z.